HW3 – Knowledge Representation and Reasoning

(Propositional Logic, CNF, First-Order Logic)

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Instructions

- Show your reasoning clearly for each question.
- You may use any standard logical equivalences.
- Unless otherwise stated, you can assume all variables range over an appropriate domain (e.g., people, courses, animals).
- Write neatly and justify non-trivial steps; unreadable work may not receive full credit.

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1) Propositional Reasoning in a Tiny Wumpus World (30 pts)

We consider a simplified Wumpus world with a **2x2 grid**. The agent only cares about avoiding **pits**. The coordinates are (1, 1), (1, 2), (2, 1), (2, 2). The agent starts at square (1, 1).

We use the following propositional symbols:

- B_{ij}: There is a breeze in square (i, j).
- P_{ij}: There is a pit in square (i, j).

We assume the usual Wumpus-world rule for breezes and pits. In particular:

$$B_{11} \leftrightarrow (P_{12} \lor P_{21}).$$

In summary, the knowledge base KB for this question consists of:

(i) There is breeze in (1,1)
(ii) $B_{11} \leftrightarrow (P_{12} \lor P_{21})$.
(iii) No other information about pits or breezes is known.

We are interested in entailment about $\alpha := P_{12}$.

(a) Truth Table (10 pts)

Fill a truth table involving propositions B_{11}, P_{11}, P_{12}, P_{21}, P_{22} and *KB*.

B11	P11	P12	P21	P22	KB
0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	1	0	0
0	0	0	1	1	0
0	0	1	0	0	0
0	0	1	0	1	0
0	0	1	1	0	0
0	0	1	1	1	0
0	1	0	0	0	0
0	1	0	0	1	0
0	1	0	1	0	0
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	0	1	0
0	1	1	1	0	0
0	1	1	1	1	0
1	0	0	0	0	0
1	0	0	0	1	0
1	0	0	1	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	0	1	0
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	0	1	1
1	1	1	1	0	1
1	1	1	1	1	1

(b) Does KB entail P_{12}? (10 pts)

Circle **one**:

KB $\models P_{12}$ $\qquad$ KB $\not\models P_{12}$

Explain your answer in one line, by referring to the truth table.

Answer: $KB \not\models P_{12}$.
Reason: There is a model where KB is true and P_{12} is false.
Example row:
$B_{11} = 1, P_{11} = 0, P_{12} = 0, P_{21} = 1, P_{22} = 1$: here **KB = 1,** $P_{12} = 0$.

(c) Does KB entail $\neg P_{12}$? (10 pts)

Circle **one**:

KB $\models \neg P_{12}$ $\qquad$ KB $\not\models \neg P_{12}$

Explain your answer in one line, by referring to the truth table.

Answer: $KB \not\models \neg P_{12}$.

Reason: There is a model where KB is true and P_{12} is true.
Example row:
$B_{11} = 1, P_{11} = 0, P_{12} = 1, P_{21} = 0, P_{22} = 0$: here **KB = 1,** $P_{12} = 1$.

So KB forces "at least one of P_{12}, P_{21}" but not which one.

2) Conjunctive Normal Form (CNF) in the Wumpus World (30 pts)

(a) Propositional sentence (5 pts)

Consider the following English sentence:

"If there is no breeze in (1, 1)*, then there is no pit in* (1, 2) *and no pit in* (2, 1)*."*

Write this sentence as a propositional formula using B_{11}, P_{12}, and P_{21}.

¬B11→(¬P12∧¬P21)

(b) CNF conversion (25 pts)

Take your formula from part (a) and convert it to **equivalent CNF**. You should:

- Show each transformation step.
- Briefly name or describe the rule used at that step (e.g., "eliminate implication", "push ¬ inward (De Morgan)", "distribute ∨ over ∧", etc.).

φ=¬B11→(¬P12∧¬P21)

Eliminate implication $A \rightarrow B \equiv \neg A \vee B$:

φ≡¬(¬B11)∨(¬P12∧¬P21)

Double negation: $\neg(\neg B_{11}) \equiv B_{11}$:

≡B11∨(¬P12∧¬P21)

Distribute ∨over ∧:

≡(B11∨¬P12)∧(B11∨¬P21)

3) First–Order Logic (FOL) Representation (40 pts)

Represent the following sentences in **First-Order Logic**. Use the most reasonable meaning for each predicate.

You may use any of the following predicates (you do not have to use all of them):

- Enrolled(*x,* CS): *x* is enrolled in the CS program.
- Takes(*x,* course): *x* takes course.
- Greedy(*x*): *x* is greedy.
- Likes(*x, y*): *x* likes *y*.
- Person(*x*): *x* is a person.
- Hardworking(*x*): *x* is hard-working.
- Takes(*x, y*): *x* takes course *y*.
- Pass(*x, y*): *x* passes course *y*.
- Halfsibling(*x, y*): *x* and *y* are half-siblings.
- Father(*x, y*): *x* is father of *y*.
- Mother(*x, y*): *x* is mother of *y*.

(a) 10 pts

"Every CS student takes CS404 or CS408."

$\forall x(Enrolled(x,CS) \rightarrow (Takes(x,CS404) \lor Takes(x,CS408)))$

(b) 10 pts

"No one likes greedy people."

$\forall x \forall y(Greedy(y) \rightarrow \neg Likes(x,y))$

(c) 10 pts

"If a hard-working student will pass a course s/he is taking."

$\forall x \forall y(Hardworking(x) \land Takes(x,y) \rightarrow Pass(x,y))$

(d) 10 pts

"Half siblings have a different mom or dad"

$\forall x \forall y(Halfsibling(x,y) \rightarrow (\exists m1 \exists m2(Mother(m1,x) \land Mother(m2,y) \land m1 \neq m2) \lor \exists f1 \exists f2(Father(f1,x) \land Father(f2,y) \land f1 \neq f2)))$